

Modal GPU Cost Comparison

1. Modal GPU Pricing (Current 2026)

GPU	VRAM	\$/hr	Per call	Speed	Cost/mo (100 calls/day)
T4	16 GB	\$0.59	\$0.002	4-8 sec	\$17.70
L4	24 GB	\$1.10	\$0.003	2-4 sec	\$33.00

2. Monthly Cost Breakdown (100 calls/day)

Component	T4 (\$0.59/hr)	L4 (\$1.10/hr)
Cost per inference	\$0.002	\$0.003
Calls per day	100	100
Daily cost	\$0.20	\$0.30
Monthly cost (30 days)	\$6.00	\$9.00
Monthly savings with T4	\$3.00	-
Annual savings with T4	\$36.00	-

3. Inference Speed Comparison

GPU	Model Load	Per Call	Total (cold start)
T4	10-15 sec	4-8 sec	15-23 sec
L4	5-8 sec	2-4 sec	7-12 sec

4. Key Insight: Cold Start vs Warm Calls

Modal containers scale to zero. Each cold start loads the model (~5-15 sec). Once warm, subsequent calls are fast (2-8 sec). For a trading bot that makes 1-5 calls per day, most calls will be cold starts.

With 100 calls/day, you pay for model load time ONCE per container lifecycle.

If containers stay warm between calls (within idle timeout), you only pay for the inference time, not repeated model loads.

5. Recommendation: T4 Is More Economical

Since speed is not the priority, T4 at \$0.59/hr is 46% cheaper than L4 at \$1.10/hr. The extra 2-4 seconds per call is negligible for a daily trading bot that runs once per day.

Monthly savings: \$3.00 (T4 vs L4)

Annual savings: \$36.00 (T4 vs L4)

Speed tradeoff: +2-4 sec per call (unnoticeable for daily trading)

6. Important Caveat

T4 has 16GB VRAM vs L4 has 24GB. Llama-3-8B in 4-bit needs ~10-12GB. Both GPUs can run the model, but T4 has less headroom for larger batch sizes or longer sequences. For single inference calls (batch=1, seq_len=1024), T4 is sufficient.